Jorge Romeu Huidobro *Computer Vision Engineer*

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Research Engineer with experience and passion in computer graphics, 3D vision, and imaging. I am particularly interested in bridging classical graphics and geometry processing with modern deep learning techniques, and enjoy working at the intersection of fundamental research and production systems.

Experience

Computer Vision Research Engineer, *ALLSIDES*
Jul 2025 – Present
Brixen, Italy · 1y 5m

Research Engineer working on developing 3D scanners and the 3D-reconstruction software stack on top of it, for e-commerse, and synthetic-data generation use-case.

- Developed a physically-based differentiable rendering pipeline for BRDF reconstruction lightstage captures
- Built in-house training/compute infrastructure and trained/deployed in-house models for PBR-material reconstruction / generation

Teaching Assistant, *TU Delft*
Nov 2023 – Jul 2025
Delft, Netherlands · 1y 9m

Guided students for various BSc and MSc Computer Science courses at TU Delft:
Head TA: *3D Computer Graphics* and *Formal Verification with Lean*.
TA: for *Computer Vision*, *Visual Data Processing*, *Deep Learning*.

Research Internship, *TU Delft Networked Systems Lab / IISc Zenlab*
Jun 2023 – Sep 2023
Delft, Netherlands · 4m

Research collaboration between TU Delft and IISc Bangalore developing a haptic robotic teleoperation system between the Netherlands and India.

- Built physical robotic system and perception stack (UR3, ROS2)
- Published at IEEE Transactions on Mobile Computing 2025

Education

MSc Computer Science, *TU Delft*
Sep 2022 – Jul 2025
Delft, Netherlands

Focus on Computer Graphics, Vision and Machine Learning.
Thesis: (9/10) Geometry-Guided Video Generation with Diffusion Feature Textures

BSc Computer Science and Engineering, *TU Delft*
Sep 2019 – Jul 2022
Delft, Netherlands

Systems Track, focus on low-level programming and system engineering
Minor: Computational Science (Numerical Methods)
Thesis: (9/10) Efficient direct lighting calculations for area lights with light portals

Skills

Programming	Python, Rust, C++ · low-level and GPU programming
Machine Learning	PyTorch · diffusion and generative models · end-to-end training and deployment
Graphics & Vision	Physically based and differentiable rendering · appearance modelling · 3D vision
Sensing & Capture	Design and calibration of multi-camera / projector capture rigs
Languages	English (native) · Spanish (native) · Italian (B2)

Publications

Utilizing Operator Intent for Haptic Teleoperation Under High Latencies,
IEEE Transactions on Mobile Computing
H.J.C. Kroep, P. Makridis, **J. Romeu Huidobro**, K. Wösten, D. Choudhary, N. Gnani, T.V. Prabhakar, S. Coppens, K. Van Berlo, R. Venkatesha Prasad